

ClassWork - Questions based on List in Python

1. Student Performance Analyzer

Problem Statement

A teacher has marks of students stored in a list.

```
marks = [78, 45, 92, 35, 88, 40, 99, 56]
```

Write a program to:

1. Display all passed students (marks ≥ 40).
2. Count the number of failed students.
3. Find the highest and lowest marks without using `max()` or `min()`.
4. Create a new list containing marks above 75.

Expected Output

Passed Students: [78, 45, 92, 88, 40, 99, 56]

Failed Count: 1

Highest Marks: 99

Lowest Marks: 35

Merit List: [78, 92, 88, 99]

2. ATM Transaction History

Problem Statement

A customer's transactions are stored as:

```
transactions = [5000, -2000, 3000, -1000, -500, 7000]
```

Positive values represent deposits and negative values represent withdrawals.

Write a program to:

1. Calculate the current balance.
2. Count total deposits and withdrawals.
3. Find the largest deposit and largest withdrawal.
4. Create separate lists for deposits and withdrawals.

Expected Output

Current Balance: 11500

Deposits: [5000, 3000, 7000]

Withdrawals: [-2000, -1000, -500]

Largest Deposit: 7000

Largest Withdrawal: -2000

3. Inventory Stock Alert System

Problem Statement

An inventory manager stores stock quantities as:

stock = [25, 5, 0, 12, 3, 18, 0, 30]

Write a program to:

1. Display products that are out of stock.
2. Display products that need restocking (quantity less than 10).
3. Count available products.
4. Create a new list containing only products with stock greater than or equal to 15.

Expected Output

Out of Stock Products: 2

Restock Required: [5, 0, 3, 0]

Available Products: 6

Healthy Stock: [25, 18, 30]

4. Consecutive Number Detector

Problem Statement

Given a list:

numbers = [4, 5, 6, 10, 11, 15, 16, 17]

Write a program to find all pairs of consecutive numbers.

Expected Output

4 and 5 are consecutive

5 and 6 are consecutive

10 and 11 are consecutive

15 and 16 are consecutive

16 and 17 are consecutive

Additional Challenge

Store all consecutive pairs in a new list.

[(4,5), (5,6), (10,11), (15,16), (16,17)]

5. Bus Seat Reservation Analysis

Problem Statement

A bus has seats represented as:

seats = [1, 0, 1, 1, 0, 0, 1, 1, 1, 0]

Where:

- 1 → Seat Booked
- 0 → Seat Available

Write a program to:

1. Count booked and available seats.
2. Find the first available seat and stop searching immediately.

3. Create a list of all available seat numbers.
4. Determine whether the bus is more than 70% occupied.

Expected Output

Booked Seats: 6

Available Seats: 4

First Available Seat: 2

Available Seat Numbers: [2, 5, 6, 10]

Bus Occupancy: 60%

Status: Not More Than 70% Occupied